

[1]: *## Part 0 - Warm-Up*

```
print("Hello, BigData!")
```

Hello, BigData!

[2]: *## Part 1 - Variables & Data Types*

```
my_name = "JOHNSON"
```

```
my_age = 24
```

```
my_gpa = 3.4
```

```
am_i_present = True
```

```
print(my_name ,type(my_name))
```

```
print(my_age ,type(my_age))
```

```
print(my_gpa ,type(my_gpa))
```

```
print(am_i_present ,type(am_i_present))
```

JOHNSON <class 'str'>

24 <class 'int'>

3.4 <class 'float'>

True <class 'bool'>

[3]: *## Exercise 1.2 - Type conversion*

```
age_text = "50"
```

```
age = int(age_text)
```

```
age = age + 1
```

```
print(age)
```

51

[4]: *## Question 1.3 - Think about it*

```
print("3" + "4")
```

This will display 34 , Because they are all a string (text) and + operator is going to join instead of adding them as a number

34

[5]: *## Part 2 - Strings - Working with Text*
Exercise 2.1 - f-strings

```
my_name = "johnson"
```

```
my_age = 24
```

```
print (f"Hello My name is {my_name}, {my_age} years old, studying at AUCA.")
```

Hello My name is johnson, 24 years old, studying at AUCA.

```
[6]: ## Exercise 2.2 – String tools

course = "Big Data Analytics"
to_upper = course.upper()
length = len(course)
sub_string = course[0:3]

print (f"Course_Name :{to_upper}")
print (f"lenght :{length}")
print (f"Sub_String :{sub_string}")
```

Course_Name :BIG DATA ANALYTICS
lenght :18
Sub_String :Big

```
[7]: ## Exercise 3.1

scores = [72, 85, 91, 64, 78]
first_number = scores[0]
last_number = scores[-1]
list_length = len(scores)
total = sum(scores)
avg = total / list_length
max_score = max(scores)
min_score = min(scores)

print (f"first_score : {first_number}")
print (f"last_score : {last_number}")
print (f"List_length : {list_length}")
print (f"total : {total}")
print (f" Average : {avg}")
print (f" max_score : {max_score}")
print (f" min_score : {min_score}")
```

first_score : 72
last_score : 78
List_length : 5
total : 390
Average : 78.0
max_score : 91
min_score : 64

```
[8]: ## Exercise 3.2 – Lists change

new_scores = scores.append(88)
scores[3] = 66

new_length = len(scores)
new_total = sum(scores)
new_avg = new_total / new_length

print (f"new_scores : {scores}")
print (f"new_avg : {new_avg}")
```

new_scores : [72, 85, 91, 66, 78, 88]
new_avg : 80.0

```
new_avg : 80.0
```

```
[9]: ## Part 4 · Dictionaries – Labeled Data
## Exercise 4.1
```

```
me = {
    "name": "johnson",
    "age": 24,
    "program": "Software Engineering",
    "district": "GASABO"
}
```

```
print(me["program"])
```

```
Software Engineering
```

```
[10]: ## Exercise 4.2 – Update and extend
## adding score
```

```
me["score"] = 87
me["age"] = me["age"] + 2
properties = me.keys()
```

```
print (f"Properties: {properties}")
```

```
print (f"New_dic: {me}")
```

```
Properties: dict_keys(['name', 'age', 'program', 'district', 'score'])
```

```
New_dic: {'name': 'johnson', 'age': 26, 'program': 'Software Engineering', 'district': 'GASABO', 'score': 87}
```

```
[12]:
```

```
## Part 5 · Class Dataset Analysis
```

```
import pandas as pd
```

```
df = pd.read_excel("week1_students.xlsx")
df
```

```
[12]:
```

	student_id	name	age	gender	program	district	attendance_pct	score
0	AUCA001	Aline Uwase	21	F	Information Systems	Gasabo	92	85
1	AUCA002	Eric Niyonzima	23	M	Software Engineering	Kicukiro	88	78
2	AUCA003	Diane Mukamana	20	F	Information Systems	Nyarugenge	95	91
3	AUCA004	Jean Bosco Habimana	24	M	Networking	Gasabo	70	64
4	AUCA005	Clarisse Ingabire	22	F	Accounting	Musanze	85	72
5	AUCA006	Patrick Mugisha	21	M	Software Engineering	Huye	90	83
6	AUCA007	Sandrine Umutoni	19	F	Information Systems	Kicukiro	97	94
7	AUCA008	Emmanuel Nsengiyumva	25	M	Networking	Rubavu	65	58
8	AUCA009	Grace Uwimana	20	F	Accounting	Gasabo	82	76
9	AUCA010	Yves Ndayisenga	22	M	Software Engineering	Nyarugenge	78	69
10	AUCA011	Josiane Mukeshimana	21	F	Information Systems	Muhanga	93	88

11	AUCA012	Claude Iradukunda	23	M	Networking	Gasabo	75	66
12	AUCA013	Beata Nyirahabimana	24	F	Accounting	Nyamagabe	89	81
13	AUCA014	Olivier Twagirimana	20	M	Software Engineering	Kicukiro	96	90
14	AUCA015	Chantal Uwera	22	F	Information Systems	Rusizi	68	61
15	AUCA016	Fabrice Byiringiro	21	M	Networking	Musanze	84	74
16	AUCA017	Solange Ishimwe	19	F	Software Engineering	Gasabo	91	87
17	AUCA018	David Hakizimana	25	M	Accounting	Nyagatare	72	55
18	AUCA019	Immaculee Uwamahoro	23	F	Information Systems	Kicukiro	87	79
19	AUCA020	Samuel Rukundo	20	M	Networking	Nyarugenge	94	92

```
[13]: students = df.to_dict("records")
students
```

```
{'district': 'Muhanga',
 'attendance_pct': 93,
 'score': 88},
{'student_id': 'AUCA012',
 'name': 'Claude Iradukunda',
 'age': 23,
 'gender': 'M',
 'program': 'Networking',
 'district': 'Gasabo',
 'attendance_pct': 75,
 'score': 66},
{'student_id': 'AUCA013',
 'name': 'Beata Nyirahabimana',
 'age': 24,
 'gender': 'F',
 'program': 'Accounting',
 'district': 'Nyamagabe',
 'attendance_pct': 89,
```

```
[14]: ## Exercise 5.1 – Describe the class
```

```
scores = [student["score"] for student in students]

print(f"Number of students :{len(students)}")
print(f"Class average :{round(sum(scores)/len(scores),1)}")
print(f"Highest score :{max(scores)}")
print(f"Lowest score :{min(scores)}")
print(f"Score_Range :{max(scores) - min(scores)}")
```

```
Number of students :20
Class average :(77, 1)
Highest score :94
Lowest score :55
Score_Range :39
```

```
[15]: ## Exercise 5.2 – Inspect individual records
```

```
print(f"First_student : {students[0]}")
print(f>Last_student : {students[-1]}")
```

```
[15]: ## Exercise 5.2 – Inspect individual records

print (f"First_student : {students[0]}")
print (f>Last_student : {students[-1]}")

First_student : {'student_id': 'AUCA001', 'name': 'Aline Uwase', 'age': 21, 'gender': 'F', 'program': 'Information Systems', 'district': 'Gasabo', 'attendance_pct': 92, 'score': 85}
Last_student : {'student_id': 'AUCA020', 'name': 'Samuel Rukundo', 'age': 20, 'gender': 'M', 'program': 'Networking', 'district': 'Nyarugenge', 'attendance_pct': 94, 'score': 92}
```

```
[16]: ## Exercise 5.3 – Count by category

student = students[6]

print(f"{student["name"]} from {student["district"]} studies {student["program"]} and scored {student["score"]}.".")

Sandrine Umutoni from Kicukiro studies Information Systems and scored 94.
```

```
[18]: ## Exercise 5.3 – Count by category

districts = [student["district"] for student in students]
print(districts)

print(f"Gasabo : {districts.count("Gasabo")}")
print(f"Kicukiro : {districts.count("Kicukiro")}")
print(f"Gasabo : {districts.count("Nyarugenge")}")

['Gasabo', 'Kicukiro', 'Nyarugenge', 'Gasabo', 'Musanze', 'Huye', 'Kicukiro', 'Rubavu', 'Gasabo', 'Nyarugenge', 'Muhanga', 'Gasabo', 'Nyamagabe', 'Kicukiro', 'Rusizi', 'Musanze', 'Gasabo', 'Nyagatare', 'Kicukiro', 'Nyarugenge']
Gasabo : 5
Kicukiro : 4
Gasabo : 3
```

```
[19]: ## Exercise 5.4 – Stretch goal: who is the top student?

names = [student["name"] for student in students]

best_score = max(scores)
position = scores.index(best_score)
best_name = names[position]

print(f"The best student is {best_name} with a score of {best_score}.".);

The best student is Sandrine Umutoni with a score of 94.
```

```
[20]: ## stretch

attendances = [student["attendance_pct"] for student in students]

lowest_score = min(scores)
position = scores.index(lowest_score)
lowest_name = names[position]

print(f"The lowest student is {lowest_name} with a score of {lowest_score} and attendance of {attendances[position]}%.".)

The lowest student is David Hakizimana with a score of 55 and attendance of 72%.
```